

HW 3: Introduction to a Significance Test

Directions: For problems 1 – 6, develop the null and alternative hypotheses and define p in words, for the following situations.

1) A company claims that 70% of its customers are satisfied with their service. A disgruntled customer believes this is an overstatement, and wants to conduct a survey to test this claim. A random sample of 200 customers is taken, and it is found that 130 of them are satisfied.

2) A university states that 60% of its students pass their exams on the first attempt. A teacher believes the actual proportion of students who pass on the first attempt is much higher. They take a sample of 150 students who take a first quarter exam, and 80 pass on their first try.

3) You state health department claims that 40% of the population exercise regularly. A local health official wants to test this claim in their town. They send out a survey to 500 residents, and it is found that 220 of them exercise regularly.

4) A manufacturer claims that only 10% of its products are defective. A quality control specialist wants to test this to make sure that proportion is not higher than the claim. In a sample of 300 products, they found 35 are found to be defective.

5) A politician claims that 55% of voters in their district support their policies. A recent smear campaign from the other political party has them worried that this support might be lowered. Their team takes a poll of 1000 voters, and find that 525 of them support the policies.

6) A technology company claims that 25% of its users upgrade to the premium version of their software after the free trial. A watchdog organization wants to investigate this claim to see if they find evidence that it is not true. In a sample of 400 users who tried the free trial, 110 upgraded to the premium version.

7) For the following questions, write T for true and F for false for the given statement. If it is false, in the space to the right, write down how it should be changed to make it true.

- _____ a) When conducting a significance test for proportions, the probability computed assuming H_0 is true.
- _____ b) If the p-value is larger than α , we reject H_0 and conclude there is convincing evidence in favor of the alternative hypothesis.
- _____ c) When the p-value is smaller than α , we always say that the result of the study is statistically significant.
- _____ d) We should always check the 10% condition to see if our independence condition is satisfied.
- _____ e) When the p-value is larger than α , the difference between $\hat{p}$ and p_0 is always attributed to sampling variability.
- _____ f) We check the large counts condition by using $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.
- _____ g) Saying “fail to reject the null hypothesis” and “accept the null hypothesis” are the same thing.
- _____ h) The probability of an event occurring by chance, if H_a is true, is always low when the result of the study is statistically significant.

8) Give the following information, find the z-score, p-value, and write a conclusion without context.

<i>Given</i>	<i>z-score</i>	<i>p-value</i>	<i>Conclusion without Context</i>
$\hat{p} = 0.4$ $p_0 = 0.25$ $n = 50$ $\alpha = 0.01$ Alternative: $>$			
$\hat{p} = 0.5$ $p_0 = 0.61$ $n = 45$ $\alpha = 0.05$ Alternative: $<$			
$\hat{p} = 0.3$ $p_0 = 0.45$ $n = 55$ $\alpha = 0.05$ Alternative: $\neq$			
$\hat{p} = 0.84$ $p_0 = 0.80$ $n = 450$ $\alpha = 0.05$ Alternative: $\neq$			